

SOLIDWORKS API (.NET, VBA, C#) — Reference

(26/337): IBody2 (part 1 of 5)

SOLIDWORKS API Documentation (converted from CHM)

IBody2 (part 1 of 5) (Type)

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IBody2 Interface

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[SolidWorks.Interop.sldworks Namespace](#) : IBody2 Interface

Allows access to the faces on a body and the ability to create surfaces for sewing into a body object.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IBody2
```

Visual Basic (Usage)

```
Dim instance As IBody2
```

C#

```
public interface IBody2
```

C++/CLI

```
public interface class IBody2
```

Visual Basic for Applications (VBA) Syntax

See [Body2](#).

Example

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 Access Diagram

[Body2](#)

 See Also

[IBody2 Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IEnumBodies2 Interface](#)

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IBody2 Interface Members

[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IBody2 Interface

The following tables list the members exposed by [IBody2](#).

 Public Properties

Name	Description
 Check3	Gets whether the body is a valid and returns an IFaultEntity object if any faults exist.
 DisableDisplay	Gets or sets whether to hide or show this body.
 DisableHighlight	Disables highlighting of the selected body in the graphics area.
 IMaterialPropertyValues	Obsolete. Superseded by IBody2::IMaterialPropertyValues2 .
 IMaterialPropertyValues2	Gets or sets the material properties for a body other than the base body in the active configuration.
 IsSafe	Not implemented.
 MaterialPropertyValues	Obsolete. Superseded by IBody2::MaterialPropertyValues2 .
 MaterialPropertyValues2	Gets or sets the material properties for a body other than the base body in the active configuration.
 Name	Gets or sets the name of the selected body.
 Visible	Gets whether this body is visible or hidden.

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 Public Methods

Name	Description
 AddConstantFillet	Creates constant radius fillets on the specified edges on this body.
 AddProfileArc	Creates an arc profile curve and returns a pointer to that curve.
 AddProfileBsplines	Creates an B-spline profile curve and returns a pointer to that curve.
 AddProfileBsplinesByPts	Adds a profile B-spline.
 AddProfileLine	Creates a line profile curve and returns a pointer to that curve.

	AddPropertyExtension	Obsolete. Superseded by iBody2::AddPropertyExtension2
	AddPropertyExtension2	Adds a property extension to this body.
	AddVertexPoint	Adds a vertex.
	ApplyTransform	Applies a transform to this body.
	Check	Obsolete. Superseded by iBody2::Check3
	Check2	Obsolete. Superseded by iBody2::Check3
	ConvertToMeshBody	Converts a solid body to a mesh BREP (boundary representation) body.
	Copy	Obsolete. Superseded by iBody2::Copy2
	Copy2	Gets a copy of this body.
	CreateBaseFeature	Creates a base feature for the imported body.
	CreateBlendSurface	Creates a constant radius rolling-ball blend surface (also known as a pipe surface) between two side surfaces.
	CreateBodyFromFaces	Creates a temporary body from the faces.
	CreateBodyFromSurfaces	Creates a body from a list of trimmed surfaces.
	CreateBoundedSurface	Creates a bounded surface from an independent base surface.
	CreateB-splineSurface	Creates a new B-spline surface.
	CreateExtrusionSurface	Creates a new surface of extrusion (infinitely long tabulated cylinder).
	CreateNewSurface	Creates a handle for a new surface to serve as geometry for a face to be added to the body.
	CreateOffsetSurface	Creates a new surface offset from an existing surface.
	CreatePlanarSurface	Creates a new infinite planar surface.
	CreatePlanarTrimSurfaceDLL	Creates a planar trim surface for this body.
	CreateRevolutionSurface	Creates a new surface of revolution.
	CreateRuledSurface	Creates a ruled surface from the specified curves and apex point.
	CreateTempBodyFromSurfaces	Creates a temporary body from a list of existing trimmed surfaces.
	CreateTrimmedSurface	Creates a trimmed surface from a base surface and a list of existing trimming curves.
	DeleteBlends	Obsolete. Superseded by iBody2::DeleteBlends2
	DeleteBlends2	Obsolete. Superseded by iBody2::DeleteBlends3
	DeleteBlends3	Removes a set of fillet faces from a temporary body and heals the body.
	DeleteFaces	Obsolete. Superseded by iBody2::DeleteFaces3
	DeleteFaces2	Obsolete. Superseded by iBody2::DeleteFaces3
	DeleteFaces3	Obsolete. Superseded by iBody2::DeleteFaces4
	DeleteFaces4	Obsolete. Superseded by iBody2::DeleteFaces5
	DeleteFaces5	Deletes a set of faces from a temporary body.
	DeleteFacesMakeSheetBodies	Creates sheet bodies from deleted faces.
	DeSelect	Deselects this body.
	Diagnose	Gets the iDiagnoseResult object for this body.
	Display	Obsolete. Superseded by iBody2::Display3
	Display2	Obsolete. Superseded by iBody2::Display3
	Display3	Displays this temporary body in the context of the specified part or component.
	DisplayWireFrameXOR	Displays a temporary body in the given part's context in XOR mode.
	DraftBody	Obsolete. Superseded by iBody2::DraftBody2
	DraftBody2	Adds drafts to the specified faces on a temporary body. This method modifies the body.
	EnumFaces	Returns an enumerated list of the faces in a body.
	ExtendSurface	Creates a new temporary body by extending the selected edges.

		
	FindAttribute	Finds an attribute on a body.
	GetBodyBox	Gets the bounding box for this body.
	GetCoincidenceTransform	Obsolete. Superseded by IBody2::GetCoincidenceTransform2
	GetCoincidenceTransform2	Calculates the transformation matrix, which would make the input body coincident with this body if the transformation matrix is applied.
	GetEdgeCount	Gets the number of edges for this body.
	GetEdges	Gets the edges for this body.
	GetExcessBodyArray	Gets the excess bodies after sewing.
	GetExtremePoint	Calculates the extreme point of the model in the specified direction.
	GetFaceCount	Gets the number of faces in this body.
	GetFaces	Gets all of the faces on the body.
	GetFeatureCount	Gets the number of features in this body in a multibody sheet metal part.
	GetFeatures	Gets the features in this body in a multibody sheet metal part.
	GetFirstFace	Finds the first face in a body and returns the face.
	GetFirstSelectedFace	Gets the first selected face on this body. For use with IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	GetGraphicsBody	Gets the graphics body associated with this body.
	GetIgesErrorCode	Gets the current IGES error code.
	GetIgesErrorCount	Gets the number of errors encountered while running an IGES routine.
	GetIntersectionEdges	Obsolete. Superseded by IBody2::GetIntersectionEdges2 .
	GetIntersectionEdges2	Gets the edges resulting from the intersection of the specified tool body and this body.
	GetMassProperties	Gets the mass properties of this body.
	GetMaterialIdName	Obsolete. Superseded by IBody2::GetMaterialIdName2
	GetMaterialIdName2	Gets the material name for this body.
	GetMaterialPropertyName	Gets the names of the material database and the material for the specified configuration.
	GetMaterialUserName	Obsolete. Superseded by IBody2::GetMaterialUserName2
	GetMaterialUserName2	Gets the material name for this body; the material name is visible to the user.
	GetMeshBody	Gets the mesh body associated with this body.
	GetMiddleSurface	Inserts a midsurface in a body.
	GetNextSelectedFace	Gets the next selected face. For use with IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	GetOriginalPatternedBody	Gets the original body from this patterned body. Also gets the transformation of this body with respect to the original body.
	GetProcessedBody	Obsolete. Superseded by IBody2::GetProcessedBody2 .
	GetProcessedBody2	Pre-processes the geometry of the body using the specified parameters.
	GetProcessedBodyWithSelfFace	Gets a processed body.
	GetPropertyExtension	Obsolete. Superseded by IBody2::GetPropertyExtension2 .
	GetPropertyExtension2	Gets the specified property extension on this body.
	GetSafeBody	Not implemented.
	GetSelectedFaceCount	Gets the number of selected faces on this body. For use with IBody2::GetProcessedBodyWithSelfFace and IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	GetSelectedFaces	Gets the selected faces. For use with IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	GetSelectionId	Gets the selection ID of the body, if one exists.
	GetSheetBody	Gets a sheet (surface) body in this body.
	GetTessellation	Gets the ITessellation object.
	GetTexture	Gets the texture applied to this body in the specified configuration.
	GetTrackingIDs	Gets the tracking IDs assigned to this body .

	GetTrackingIdsCount	Gets the number of tracking IDs on this body.
	GetType	Gets the type of the body.
	GetUpdateStamp	Gets the update stamp for the body as of the last rebuild.
	GetVertexCount	Gets the number of vertices in this body.
	GetVertices	Gets the vertices in this body.
	HasMaterialPropertyValues	Gets whether this body has an appearance.
	Hide	Hides this temporary body in the context of the specified part.
	HideBody	Hides or shows this body.
	IAddProfileArc	Creates an arc profile curve and returns a pointer to that curve.
	IAddProfileArcDLL	Adds a profile arc.
	IAddProfileB spline	Creates an B-spline profile curve and returns a pointer to that curve.
	IAddProfileB splineByPts	Adds a profile B-spline.
	IAddProfileB splineDLL	Adds a profile B-spline.
	IAddProfileLine	Creates a line profile curve and returns a pointer to that curve.
	IAddProfileLineDLL	Adds a profile line.
	IAddVertexPoint	Adds a vertex.
	ICombineVolumes	Combines the volumes of two bodies.
	ICopy	Gets a copy of this body.
	ICreateBaseFeature	Creates a base feature for the imported body.
	ICreateBlendSurface	Creates a constant radius rolling-ball blend surface (also known as a pipe surface) between two side surfaces.
	ICreateBodyFromFaces	Creates a temporary body from the faces.
	ICreateBoundedSurface	Creates a bounded surface from an independent base surface.
	ICreateB splineSurface	Creates a new B-spline surface.
	ICreateB splineSurfaceDLL	Creates a B-spline surface in this body.
	ICreateExtrusionSurface	Creates a new surface of extrusion (infinitely long tabulated cylinder).
	ICreateExtrusionSurfaceDLL	Creates an extruded surface.
	ICreateNewSurface	Creates a handle for a new surface to serve as geometry for a face to be added to the body.
	ICreateOffsetSurface	Creates a new surface offset from an existing surface.
	ICreatePlanarSurface	Creates a new infinite planar surface.
	ICreatePlanarSurfaceDLL	Creates a planar surface.
	ICreatePlanarTrimSurfaceDLL	Creates a planar trim surface for this body.
	ICreateP splineSurfaceDLL	Creates a B-surface from a piecewise surface.
	ICreateRevolutionSurface	Creates a new surface of revolution.
	ICreateRevolutionSurfaceDLL	Creates a surface of revolution for this body.
	ICreateRuledSurface	Creates a ruled surface from the specified curves and apex point.
	ICreateTempBodyFromSurfaces	Creates a temporary body from a list of existing trimmed surfaces.
	IDeleteBlends	Obsolete. Superseded by IBody2::IDeleteBlends2 .
	IDeleteBlends2	Obsolete. Superseded by IBody2::DeleteBlends3 .
	IDeleteBlends3	Removes a set of fillet faces from a temporary body and heals the body.
	IDeleteFaces2	Obsolete. Superseded by IBody2::IDeleteFaces3 .
	IDeleteFaces3	Obsolete. Superseded by IBody2::IDeleteFaces4 .
	IDeleteFacesMakeSheetBodies	Creates sheet bodies from deleted faces.

		
	IDeleteFacesMakeSheetBodiesCount	Gets the number of sheet bodies to create from the deleted faces.
	IDisplay	Obsolete. Superseded by IBody2::Display3 .
	IDisplayWireFrameXOR	Displays a temporary body in the given part's context in XOR mode.
	IDraftBody	Obsolete. Superseded by IBody2::IDraftBody2 .
	IDraftBody2	Adds drafts to the specified faces on a temporary body. This method modifies the body.
	IExtendSurface	Creates a new temporary body by extending the selected edges.
	IGetBodyBox	Gets the bounding box for this body.
	IGetEdges	Gets the edges for this body.
	IGetExcessBodyArray	Gets the excess bodies after sewing.
	IGetExcessBodyCount	Gets the number of excess bodies.
	IGetFaces	Gets all of the faces on the body.
	IGetFeatures	Gets the features in this body in a multibody sheet metal part.
	IGetFirstFace	Finds the first face in a body and returns the face.
	IGetFirstSelectedFace	Gets the first selected face on this body. For use with IBody2::IGetProcessedBodyWithSelfFace and intended for IGES routines.
	IGetIntersectionEdgeCount	Gets the number of intersecting edges between this body and the specified tool body.
	IGetIntersectionEdges	Obsolete. Superseded by IBody2::GetIntersectionEdges2 .
	IGetMassProperties	Gets the mass properties of this body.
	IGetMaterialPropertyValuesForFace	Gets the color of the specified face.
	IGetMiddleSurface	Inserts a midsurface in a body.
	IGetNextSelectedFace	Gets the next selected face. For use with IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	IGetProcessedBody	Obsolete. Superseded by IBody2::GetProcessedBody2 .
	IGetProcessedBodyWithSelfFace	Gets a processed body.
	IGetSelectedFaces	Gets the selected faces. For use with IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	IGetSheetBody	Gets a sheet (surface) body in this body.
	IGetTessellation	Gets the ITessellation object.
	IGetTrackingIDs	Gets the tracking IDs assigned to this body .
	IGetVertices	Gets the vertices in this body.
	IHide	Hides a temporary body using the specified part's context.
	IMatchedBoolean	Obsolete. Superseded by IBody2::IMatchedBoolean3 .
	IMatchedBoolean2	Obsolete. Superseded by IBody2::IMatchedBoolean3 .
	IMatchedBoolean3	Obsolete. Superseded by IBody2::IMatchedBoolean4 .
	IMatchedBoolean4	Performs a matched boolean on the specified bodies and supports an optional list of faces that match exactly.
	IOperations	Obsolete. Superseded by IBody2::Operations2 .
	IOperations2	Performs add, cut, and intersect (unite, subtract, and interfere) operations between two temporary bodies.
	IRemoveFacesFromSheet	Removes the specified faces from a sheet (surface) body.
	IRemoveMaterialProperty	Removes the material property values (e.g., color) from the body in the specified configurations in parts and assemblies.
	ISave	Saves this body.
	ISectionBySheet	Sections a body using a sheet (surface) body.
	ISetCurrentSurface	Places an existing surface into a temporary body that is under construction.
	ISetXform	Obsolete. Superseded by IBody2::ApplyTransform .
	IsGraphicsBody	Gets whether this body is a graphics body.
	IsMeshBody	Gets whether this body is a mesh body.

		
	IsPatternSeed	Gets whether this body is the seed of a patterned body.
	IsSheetMetal	Gets whether this body was created by a sheet metal feature.
	IsTemporaryBody	Gets whether the body is a temporary body.
	MakeOffset	Creates a new temporary body by offsetting the selected surface body by the specified distance and in the specified direction.
	MatchedBoolean	Obsolete. Superseded by IBody2::MatchedBoolean3 .
	MatchedBoolean2	Obsolete. Superseded by IBody2::MatchedBoolean3 .
	MatchedBoolean3	Obsolete. Superseded by IBody2::MatchedBoolean4 .
	MatchedBoolean4	Performs a matched boolean on the specified bodies and supports an optional list of faces that match exactly.
	MinimumRadius	Gets the minimum radius of this body.
	Negate	Reverses the direction (i.e., orientation) of the body.
	OffsetPlanarWireBody	Offsets a planar wire body in the normal plane by the specified distance.
	Operations	Obsolete. Superseded by IBody2::Operations2 .
	Operations2	Performs add, cut, and intersect (unite, subtract, and interfere) operations between two temporary bodies.
	RemoveFacesFromSheet	Removes the specified faces from a sheet (surface) body.
	RemoveMaterialProperty	Removes the material property values (e.g., color) from the body in the specified configurations in parts and assemblies.
	RemoveRedundantTopology	Removes redundant topology from the body.
	RemoveTexture	Removes the texture applied to this body in the specified configuration.
	RemoveTextureByDisplayState	Removes the texture applied to this body in the specified display state.
	RemoveTrackingID	Removes a tracking ID assigned to this body .
	ResetEdgeTolerances	Resets the tolerance on all edges of this body.
	ResetPropertyExtension	Obsolete. Superseded by IBody2::ResetPropertyExtension2 .
	ResetPropertyExtension2	Clears all values stored in the property extension.
	Save	Saves this body.
	Select	Obsolete. Superseded by IBody2::Select2 .
	Select2	Selects this body and marks it.
	SetCurrentSurface	Places an existing surface into a temporary body that is under construction.
	SetIgesInfo	Sends IGES-specific data to the geometric modeler.
	SetMaterialIdName	Obsolete. Superseded by IBody2::SetMaterialIdName2 .
	SetMaterialIdName2	Sets the material name for this body.
	SetMaterialProperty	Sets the material for this body.
	SetMaterialUserName	Obsolete. Superseded by IBody2::SetMaterialUserName2 .
	SetMaterialUserName2	Sets the material name for this body. This material name is visible to the user.
	SetTexture	Applies texture to this body in the specified configuration.
	SetTextureByDisplayState	Sets the texture applied to this body in the specified display state.
	SetTrackingID	Assigns a tracking ID to this body.
	SetXform	Obsolete. Superseded by IBody2::ApplyTransform .

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See Also

[IBody2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IEnumBodies2 Interface](#)

IBody2 Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IBody2 Interface

For a list of all members of this type, see [IBody2 members](#).

Public Methods

Name	Description
AddConstantFillet	Creates constant radius fillets on the specified edges on this body.
AddProfileArc	Creates an arc profile curve and returns a pointer to that curve.
AddProfileB-spline	Creates an B-spline profile curve and returns a pointer to that curve.
AddProfileB-splineByPts	Adds a profile B-spline.
AddProfileLine	Creates a line profile curve and returns a pointer to that curve.
AddPropertyExtension	Obsolete. Superseded by IBody2::AddPropertyExtension2 .
AddPropertyExtension2	Adds a property extension to this body.
AddVertexPoint	Adds a vertex.
ApplyTransform	Applies a transform to this body.
Check	Obsolete. Superseded by IBody2::Check3 .
Check2	Obsolete. Superseded by IBody2::Check3 .
ConvertToMeshBody	Converts a solid body to a mesh BREP (boundary representation) body.
Copy	Obsolete. Superseded by IBody2::Copy2 .
Copy2	Gets a copy of this body.
CreateBaseFeature	Creates a base feature for the imported body.
CreateBlendSurface	Creates a constant radius rolling-ball blend surface (also known as a pipe surface) between two side surfaces.
CreateBodyFromFaces	Creates a temporary body from the faces.
CreateBodyFromSurfaces	Creates a body from a list of trimmed surfaces.
CreateBoundedSurface	Creates a bounded surface from an independent base surface.
CreateB-splineSurface	Creates a new B-spline surface.
CreateExtrusionSurface	Creates a new surface of extrusion (infinitely long tabulated cylinder).
CreateNewSurface	Creates a handle for a new surface to serve as geometry for a face to be added to the body.
CreateOffsetSurface	Creates a new surface offset from an existing surface.
CreatePlanarSurface	Creates a new infinite planar surface.
CreatePlanarTrimSurfaceDLL	Creates a planar trim surface for this body.
CreateRevolutionSurface	Creates a new surface of revolution.
CreateRuledSurface	Creates a ruled surface from the specified curves and apex point.
CreateTempBodyFromSurfaces	Creates a temporary body from a list of existing trimmed surfaces.
CreateTrimmedSurface	Creates a trimmed surface from a base surface and a list of existing trimming curves.
DeleteBlends	Obsolete. Superseded by IBody2::DeleteBlends2 .
DeleteBlends2	Obsolete. Superseded by IBody2::DeleteBlends3 .
DeleteBlends3	Removes a set of fillet faces from a temporary body and heals the body.
DeleteFaces	Obsolete. Superseded by IBody2::DeleteFaces3 .
DeleteFaces2	Obsolete. Superseded by IBody2::DeleteFaces3 .
DeleteFaces3	Obsolete. Superseded by IBody2::DeleteFaces4 .
DeleteFaces4	Obsolete. Superseded by IBody2::DeleteFaces5 .
DeleteFaces5	Deletes a set of faces from a temporary body.

		
	DeleteFacesMakeSheetBodies	Creates sheet bodies from deleted faces.
	DeSelect	Deselects this body.
	Diagnose	Gets the IDiagnoseResult object for this body.
	Display	Obsolete. Superseded by IBody2::Display3 .
	Display2	Obsolete. Superseded by IBody2::Display3 .
	Display3	Displays this temporary body in the context of the specified part or component.
	DisplayWireFrameXOR	Displays a temporary body in the given part's context in XOR mode.
	DraftBody	Obsolete. Superseded by IBody2::DraftBody2 .
	DraftBody2	Adds drafts to the specified faces on a temporary body. This method modifies the body.
	EnumFaces	Returns an enumerated list of the faces in a body.
	ExtendSurface	Creates a new temporary body by extending the selected edges.
	FindAttribute	Finds an attribute on a body.
	GetBodyBox	Gets the bounding box for this body.
	GetCoincidenceTransform	Obsolete. Superseded by IBody2::GetCoincidenceTransform2 .
	GetCoincidenceTransform2	Calculates the transformation matrix, which would make the input body coincident with this body if the transformation matrix is applied.
	GetEdgeCount	Gets the number of edges for this body.
	GetEdges	Gets the edges for this body.
	GetExcessBodyArray	Gets the excess bodies after sewing.
	GetExtremePoint	Calculates the extreme point of the model in the specified direction.
	GetFaceCount	Gets the number of faces in this body.
	GetFaces	Gets all of the faces on the body.
	GetFeatureCount	Gets the number of features in this body in a multibody sheet metal part.
	GetFeatures	Gets the features in this body in a multibody sheet metal part.
	GetFirstFace	Finds the first face in a body and returns the face.
	GetFirstSelectedFace	Gets the first selected face on this body. For use with IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	GetGraphicsBody	Gets the graphics body associated with this body.
	GetIgesErrorCode	Gets the current IGES error code.
	GetIgesErrorCount	Gets the number of errors encountered while running an IGES routine.
	GetIntersectionEdges	Obsolete. Superseded by IBody2::GetIntersectionEdges2 .
	GetIntersectionEdges2	Gets the edges resulting from the intersection of the specified tool body and this body.
	GetMassProperties	Gets the mass properties of this body.
	GetMaterialIdName	Obsolete. Superseded by IBody2::GetMaterialIdName2 .
	GetMaterialIdName2	Gets the material name for this body.
	GetMaterialPropertyName	Gets the names of the material database and the material for the specified configuration.
	GetMaterialUserName	Obsolete. Superseded by IBody2::GetMaterialUserName2 .
	GetMaterialUserName2	Gets the material name for this body; the material name is visible to the user.
	GetMeshBody	Gets the mesh body associated with this body.
	GetMiddleSurface	Inserts a midsurface in a body.
	GetNextSelectedFace	Gets the next selected face. For use with IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	GetOriginalPatternedBody	Gets the original body from this patterned body. Also gets the transformation of this body with respect to the original body.
	GetProcessedBody	Obsolete. Superseded by IBody2::GetProcessedBody2 .
	GetProcessedBody2	Pre-processes the geometry of the body using the specified parameters.

		
	GetProcessedBodyWithSelfFace	Gets a processed body.
	GetPropertyExtension	Obsolete. Superseded by IBody2::GetPropertyExtension2 .
	GetPropertyExtension2	Gets the specified property extension on this body.
	GetSafeBody	Not implemented.
	GetSelectedFaceCount	Gets the number of selected faces on this body. For use with IBody2::GetProcessedBodyWithSelfFace and IBody2::!GetProcoessedBodyWithSelfFace and intended for IGES routines.
	GetSelectedFaces	Gets the selected faces. For use with IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	GetSelectionId	Gets the selection ID of the body, if one exists.
	GetSheetBody	Gets a sheet (surface) body in this body.
	GetTessellation	Gets the ITessellation object.
	GetTexture	Gets the texture applied to this body in the specified configuration.
	GetTrackingIDs	Gets the tracking IDs assigned to this body .
	GetTrackingIDsCount	Gets the number of tracking IDs on this body.
	GetType	Gets the type of the body.
	GetUpdateStamp	Gets the update stamp for the body as of the last rebuild.
	GetVertexCount	Gets the number of vertices in this body.
	GetVertices	Gets the vertices in this body.
	HasMaterialPropertyValues	Gets whether this body has an appearance.
	Hide	Hides this temporary body in the context of the specified part.
	HideBody	Hides or shows this body.
	IAddProfileArc	Creates an arc profile curve and returns a pointer to that curve.
	IAddProfileArcDLL	Adds a profile arc.
	IAddProfileB spline	Creates an B-spline profile curve and returns a pointer to that curve.
	IAddProfileB splineByPts	Adds a profile B-spline.
	IAddProfileB splineDLL	Adds a profile B-spline.
	IAddProfileLine	Creates a line profile curve and returns a pointer to that curve.
	IAddProfileLineDLL	Adds a profile line.
	IAddVertexPoint	Adds a vertex.
	ICombineVolumes	Combines the volumes of two bodies.
	ICopy	Gets a copy of this body.
	ICreateBaseFeature	Creates a base feature for the imported body.
	ICreateBlendSurface	Creates a constant radius rolling-ball blend surface (also known as a pipe surface) between two side surfaces.
	ICreateBodyFromFaces	Creates a temporary body from the faces.
	ICreateBoundedSurface	Creates a bounded surface from an independent base surface.
	ICreateB splineSurface	Creates a new B-spline surface.
	ICreateB splineSurfaceDLL	Creates a B-spline surface in this body.
	ICreateExtrusionSurface	Creates a new surface of extrusion (infinitely long tabulated cylinder).
	ICreateExtrusionSurfaceDLL	Creates an extruded surface.
	ICreateNewSurface	Creates a handle for a new surface to serve as geometry for a face to be added to the body.
	ICreateOffsetSurface	Creates a new surface offset from an existing surface.
	ICreatePlanarSurface	Creates a new infinite planar surface.
	ICreatePlanarSurfaceDLL	Creates a planar surface.
	ICreatePlanarTrimSurfaceDLL	Creates a planar trim surface for this body.

		
	ICreatePspineSurfaceDLL	Creates a B-surface from a piecewise surface.
	ICreateRevolutionSurface	Creates a new surface of revolution.
	ICreateRevolutionSurfaceDLL	Creates a surface of revolution for this body.
	ICreateRuledSurface	Creates a ruled surface from the specified curves and apex point.
	ICreateTempBodyFromSurfaces	Creates a temporary body from a list of existing trimmed surfaces.
	IDeleteBlends	Obsolete. Superseded by IBody2::IDeleteBlends2 .
	IDeleteBlends2	Obsolete. Superseded by IBody2::DeleteBlends3 .
	IDeleteBlends3	Removes a set of fillet faces from a temporary body and heals the body.
	IDeleteFaces2	Obsolete. Superseded by IBody2::IDeleteFaces3 .
	IDeleteFaces3	Obsolete. Superseded by IBody2::IDeleteFaces4 .
	IDeleteFacesMakeSheetBodies	Creates sheet bodies from deleted faces.
	IDeleteFacesMakeSheetBodiesCount	Gets the number of sheet bodies to create from the deleted faces.
	IDisplay	Obsolete. Superseded by IBody2::Display3 .
	IDisplayWireFrameXOR	Displays a temporary body in the given part's context in XOR mode.
	IDraftBody	Obsolete. Superseded by IBody2::IDraftBody2 .
	IDraftBody2	Adds drafts to the specified faces on a temporary body. This method modifies the body.
	IExtendSurface	Creates a new temporary body by extending the selected edges.
	IGetBodyBox	Gets the bounding box for this body.
	IGetEdges	Gets the edges for this body.
	IGetExcessBodyArray	Gets the excess bodies after sewing.
	IGetExcessBodyCount	Gets the number of excess bodies.
	IGetFaces	Gets all of the faces on the body.
	IGetFeatures	Gets the features in this body in a multibody sheet metal part.
	IGetFirstFace	Finds the first face in a body and returns the face.
	IGetFirstSelectedFace	Gets the first selected face on this body. For use with IBody2::IGetProcessedBodyWithSelfFace and intended for IGES routines.
	IGetIntersectionEdgeCount	Gets the number of intersecting edges between this body and the specified tool body.
	IGetIntersectionEdges	Obsolete. Superseded by IBody2::GetIntersectionEdges2 .
	IGetMassProperties	Gets the mass properties of this body.
	IGetMaterialPropertyValuesForFace	Gets the color of the specified face.
	IGetMiddleSurface	Inserts a midsurface in a body.
	IGetNextSelectedFace	Gets the next selected face. For use with IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	IGetProcessedBody	Obsolete. Superseded by IBody2::GetProcessedBody2 .
	IGetProcessedBodyWithSelfFace	Gets a processed body.
	IGetSelectedFaces	Gets the selected faces. For use with IBody2::GetProcessedBodyWithSelfFace and intended for IGES routines.
	IGetSheetBody	Gets a sheet (surface) body in this body.
	IGetTessellation	Gets the ITessellation object.
	IGetTrackingIDs	Gets the tracking IDs assigned to this body .
	IGetVertices	Gets the vertices in this body.
	IHide	Hides a temporary body using the specified part's context.
	IMatchedBoolean	Obsolete. Superseded by IBody2::IMatchedBoolean3 .
	IMatchedBoolean2	Obsolete. Superseded by IBody2::IMatchedBoolean3 .
	IMatchedBoolean3	Obsolete. Superseded by IBody2::IMatchedBoolean4 .

		
	IMatchedBoolean4	Performs a matched boolean on the specified bodies and supports an optional list of faces that match exactly.
	IOperations	Obsolete. Superseded by IBody2::Operations2
	IOperations2	Performs add, cut, and intersect (unite, subtract, and interfere) operations between two temporary bodies.
	IRemoveFacesFromSheet	Removes the specified faces from a sheet (surface) body.
	IRemoveMaterialProperty	Removes the material property values (e.g., color) from the body in the specified configurations in parts and assemblies.
	ISave	Saves this body.
	ISectionBySheet	Sections a body using a sheet (surface) body.
	ISetCurrentSurface	Places an existing surface into a temporary body that is under construction.
	ISetXform	Obsolete. Superseded by IBody2::ApplyTransform
	IsGraphicsBody	Gets whether this body is a graphics body.
	IsMeshBody	Gets whether this body is a mesh body.
	IsPatternSeed	Gets whether this body is the seed of a patterned body.
	IsSheetMetal	Gets whether this body was created by a sheet metal feature.
	IsTemporaryBody	Gets whether the body is a temporary body.
	MakeOffset	Creates a new temporary body by offsetting the selected surface body by the specified distance and in the specified direction.
	MatchedBoolean	Obsolete. Superseded by IBody2::MatchedBoolean3
	MatchedBoolean2	Obsolete. Superseded by IBody2::MatchedBoolean3
	MatchedBoolean3	Obsolete. Superseded by IBody2::MatchedBoolean4
	MatchedBoolean4	Performs a matched boolean on the specified bodies and supports an optional list of faces that match exactly.
	MinimumRadius	Gets the minimum radius of this body.
	Negate	Reverses the direction (i.e., orientation) of the body.
	OffsetPlanarWireBody	Offsets a planar wire body in the normal plane by the specified distance.
	Operations	Obsolete. Superseded by IBody2::Operations2
	Operations2	Performs add, cut, and intersect (unite, subtract, and interfere) operations between two temporary bodies.
	RemoveFacesFromSheet	Removes the specified faces from a sheet (surface) body.
	RemoveMaterialProperty	Removes the material property values (e.g., color) from the body in the specified configurations in parts and assemblies.
	RemoveRedundantTopology	Removes redundant topology from the body.
	RemoveTexture	Removes the texture applied to this body in the specified configuration.
	RemoveTextureByDisplayState	Removes the texture applied to this body in the specified display state.
	RemoveTrackingID	Removes a tracking ID assigned to this body .
	ResetEdgeTolerances	Resets the tolerance on all edges of this body.
	ResetPropertyExtension	Obsolete. Superseded by IBody2::ResetPropertyExtension2
	ResetPropertyExtension2	Clears all values stored in the property extension.
	Save	Saves this body.
	Select	Obsolete. Superseded by IBody2::Select2
	Select2	Selects this body and marks it.
	SetCurrentSurface	Places an existing surface into a temporary body that is under construction.
	SetIgesInfo	Sends IGES-specific data to the geometric modeler.
	SetMaterialIdName	Obsolete. Superseded by IBody2::SetMaterialIdName2
	SetMaterialIdName2	Sets the material name for this body.
	SetMaterialProperty	Sets the material for this body.
	SetMaterialUserName	Obsolete. Superseded by IBody2::SetMaterialUserName2

		
	SetMaterialUserName2	Sets the material name for this body. This material name is visible to the user.
	SetTexture	Applies texture to this body in the specified configuration.
	SetTextureByDisplayState	Sets the texture applied to this body in the specified display state.
	SetTrackingID	Assigns a tracking ID to this body.
	SetXform	Obsolete. Superseded by IBody2.ApplyTransform .

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See Also

[IBody2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IEnumerableBodies2 Interface](#)

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[Send comments on this topic.](#)

IBody2 Interface Properties

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IBody2 Interface

For a list of all members of this type, see [IBody2 members](#).

Public Properties

Name	Description
 Check3	Gets whether the body is a valid and returns an IFaultEntity object if any faults exist.
 DisableDisplay	Gets or sets whether to hide or show this body.
 DisableHighlight	Disables highlighting of the selected body in the graphics area.
 IMaterialPropertyValues	Obsolete. Superseded by IBody2.IMaterialPropertyValues2 .
 IMaterialPropertyValues2	Gets or sets the material properties for a body other than the base body in the active configuration.
 IsSafe	Not implemented.
 MaterialPropertyValues	Obsolete. Superseded by IBody2.MaterialPropertyValues2 .
 MaterialPropertyValues2	Gets or sets the material properties for a body other than the base body in the active configuration.
 Name	Gets or sets the name of the selected body.
 Visible	Gets whether this body is visible or hidden.

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See Also

[IBody2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IEnumerableBodies2 Interface](#)

SOLIDWORKS API Help

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AddConstantFillet Method (IBody2)

[See Also](#)

  [SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : AddConstantFillet Method (IBody2)

Creates constant radius fillets on the specified edges on this body.

.NET Syntax

Visual Basic (Declaration)

```
Function AddConstantFillet( _  
    ByVal Radius As System.Double, _  
    ByVal EdgesToFillet As System.Object _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IBody2
Dim Radius As System.Double
Dim EdgesToFillet As System.Object
Dim value As System.Object

value = instance.AddConstantFilletts(Radius, EdgesToFillet)
```

C#

```
System.object AddConstantFilletts(
    System.double Radius,
    System.object EdgesToFillet
)
```

C++/CLI

```
System.Object^ AddConstantFilletts(
    & System.double Radius,
    & System.Object^ EdgesToFillet
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Radius
Fillet radius
EdgesToFillet
Array of [edges](#) to fillet

Return Value

Array of newly created [faces](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::AddConstantFilletts](#).

Remarks

This method does not work for sheet bodies.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments](#) on this topic.

AddProfileArc Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : AddProfileArc Method (IBody2)

Creates an arc profile curve and returns a pointer to that curve.

.NET Syntax

Visual Basic (Declaration)

```
Function AddProfileArc( _
    ByVal Center As System.Object, _
    ByVal Axis As System.Object, _
    ByVal Radius As System.Double, _
    ByVal StartPoint As System.Object, _
```

```
ByVal EndPoint As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Center As System.Object  
Dim Axis As System.Object  
Dim Radius As System.Double  
Dim StartPoint As System.Object  
Dim EndPoint As System.Object  
Dim value As System.Object  
  
value = instance.AddProfileArc(Center, Axis, Radius, StartPoint, EndPoint)
```

C#

```
System.object AddProfileArc(  
    System.object Center,  
    System.object Axis,  
    System.double Radius,  
    System.object StartPoint,  
    System.object EndPoint  
)
```

C++/CLI

```
System.Object^ AddProfileArc(  
& System.Object^ Center,  
& System.Object^ Axis,  
& System.double Radius,  
& System.Object^ StartPoint,  
& System.Object^ EndPoint  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center
Array of 3 doubles (x,y,z)
Axis
Array of 3 doubles (x,y,z)
Radius
Arc radius
StartPoint
Array of 3 doubles (x,y,z)
EndPoint
Array of 3 doubles (x,y,z)

Return Value

Object for the curve

Visual Basic for Applications (VBA) Syntax

See [Body2::AddProfileArc](#).

Remarks

This method always creates a full circle.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::AddProfileArc Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

AddProfileBspline Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : AddProfileBspline Method (IBody2)

Creates an B-spline profile curve and returns a pointer to that curve.

.NET Syntax

Visual Basic (Declaration)

```
Function AddProfileBspline( _  
    ByVal Props As System.Object, _  
    ByVal Knots As System.Object, _  
    ByVal CtrlPtCoords As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Props As System.Object  
Dim Knots As System.Object  
Dim CtrlPtCoords As System.Object  
Dim value As System.Object  
  
value = instance.AddProfileBspline(Props, Knots, CtrlPtCoords)
```

C#

```
System.object AddProfileBspline(  
    System.object Props,  
    System.object Knots,  
    System.object CtrlPtCoords  
)
```

C++/CLI

```
System.Object^ AddProfileBspline(  
& System.Object^ Props,  
& System.Object^ Knots,  
& System.Object^ CtrlPtCoords  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Props

Contains 4 integers packed into 2 double elements (see **Remarks**)

Knots

Array of numKnots (see **Remarks**)

CtrlPtCoords

Array of numCtrlPtCoord (see **Remarks**)

Return Value

[Curve](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::AddProfileBspline](#).

Example

[Create Reference Curve \(C#\)](#)
[Create Reference Curve \(VB.NET\)](#)
[Create Reference Curve \(VBA\)](#)

Remarks

You can use this method with [IBody2::CreateRevolutionSurface](#) to generate any surface of revolution or with [IBody2::CreateExtrusionSurface](#) to generate a tabulated cylinder.

The Props argument contains the following values:

- DimensionControlPoints
- Order
- NumCtrlPoints
- Periodicity

The length of the knots array is given by:

numKnots = NumCtrlPoints + Order

The length of the CtrlPtCoords is given by:

NumCtrlPtCoord = NumCtrlPoints + DimensionControlPoints

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::AddProfileBspline Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddProfileBsplineByPts Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : AddProfileBsplineByPts Method (IBody2)

Adds a profile B-spline.

.NET Syntax

Visual Basic (Declaration)

```
Function AddProfileBsplineByPts( _  
    ByVal NumPoints As System.Integer, _  
    ByVal PointArray As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim NumPoints As System.Integer  
Dim PointArray As System.Object  
Dim value As System.Object  
  
value = instance.AddProfileBsplineByPts(NumPoints, PointArray)
```

C#

```
System.object AddProfileBsplineByPts(  
    System.int NumPoints,  
    System.object PointArray  
)
```

C++/CLI

```
System.Object^ AddProfileBsplineByPts(  
& System.int NumPoints,  
& System.Object^ PointArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumPoints

Number of B-spline points

PointArray

0-based array of 3 * NumPoints doubles

Return Value

Pointer to the profile B-spline, [ICurve](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::AddProfileBsplineByPts](#).

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

[IBody2::AddProfileBsplineByPts Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddProfileLine Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : AddProfileLine Method (IBody2)

Creates a line profile curve and returns a pointer to that curve.

.NET Syntax

Visual Basic (Declaration)

```
Function AddProfileLine( _  
    ByVal RootPoint As System.Object, _  
    ByVal Direction As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim RootPoint As System.Object  
Dim Direction As System.Object  
Dim value As System.Object  
  
value = instance.AddProfileLine(RootPoint, Direction)
```

C#

```
System.object AddProfileLine(  
    System.object RootPoint,  
    System.object Direction  
)
```

C++/CLI

```
System.Object^ AddProfileLine(  
& System.Object^ RootPoint,  
& System.Object^ Direction  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

RootPoint

Array of 3 doubles (x,y,z)

Direction

Array of 3 doubles (x,y,z)

Return Value

Object for the line profile curve

Visual Basic for Applications (VBA) Syntax

See [Body2::AddProfileLine](#).

Remarks

You can use this method with [IBody2::CreateRevolutionSurface](#) to generate a cylindrical surface of revolution or with [IBody2::CreateExtrusionSurface](#) to generate a tabulated cylinder.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::AddProfileLine Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

AddPropertyExtension Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : AddPropertyExtension Method (IBody2)

Obsolete. Superseded by [IBody2::AddPropertyExtension2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddPropertyExtension( _  
    ByVal PropertyExtension As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim PropertyExtension As System.Object  
Dim value As System.Integer  
  
value = instance.AddPropertyExtension(PropertyExtension)
```

C#

```
System.int AddPropertyExtension(  
    System.object PropertyExtension  
)
```

C++/CLI

```
System.int AddPropertyExtension(  
& System.Object^ PropertyExtension  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropertyExtension

Visual Basic for Applications (VBA) Syntax

See [Body2::AddPropertyExtension](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

AddPropertyExtension2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : AddPropertyExtension2 Method (IBody2)

Adds a property extension to this body.

.NET Syntax

Visual Basic (Declaration)

```
Function AddPropertyExtension2( _  
    ByVal PropertyExtension As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim PropertyExtension As System.Object  
Dim value As System.Integer  
  
value = instance.AddPropertyExtension2(PropertyExtension)
```

C#

```
System.int AddPropertyExtension2(  
    System.object PropertyExtension  
)
```

C++/CLI

```
System.int AddPropertyExtension2(  
& System.Object^ PropertyExtension  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropertyExtension

Value of the property extension to add to this body (see **Remarks**)

Return Value

Size of the array to which the property extension value is added

Visual Basic for Applications (VBA) Syntax

See [Body2::AddPropertyExtension2](#).

Remarks

To use this method:

1. Declare the variable.
2. Assign the variable a value: float, integer, or string.
3. Call this method to add the value to the body.

The 1-based array is a first-in-last-out structured list, whose size is used by [IBody2::GetPropertyExtension2](#) to access the property extension.

NOTE: SOLIDWORKS recommends that you use the [IAttribute](#), [IAttributeDef](#), and [IParameter](#) interfaces instead of this method. These interface provide more flexibility.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ResetPropertyExtension2 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

AddVertexPoint Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : AddVertexPoint Method (IBody2)

Adds a vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function AddVertexPoint(  
    ByVal Point As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Point As System.Object  
Dim value As System.Object  
  
value = instance.AddVertexPoint(Point)
```

C#

```
System.object AddVertexPoint(  
    System.object Point  
)
```

C++/CLI

```
System.Object^ AddVertexPoint(  
& System.Object^ Point  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Point
Vertex to add

Return Value

[IVertex](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::AddVertexPoint](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IAddVertexPoint Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ApplyTransform Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : ApplyTransform Method (IBody2)

Applies a transform to this body.

.NET Syntax

Visual Basic (Declaration)

```
Function ApplyTransform( _  
    ByVal Xform As MathTransform _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Xform As MathTransform  
Dim value As System.Boolean  
  
value = instance.ApplyTransform(Xform)
```

C#

```
System.bool ApplyTransform(  
    MathTransform Xform  
)
```

C++/CLI

```
System.bool ApplyTransform(  
& MathTransform^ Xform  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Xform

[Transform](#) to apply

Return Value

True if the transform is successfully applied, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::ApplyTransform](#).

Example

[Check Interference \(VBA\)](#)
[Combine Assembly Components into Part \(VBA\)](#)
[Cut Body in Half using Macro Feature \(VBA\)](#)

Remarks

By design, a temporary body is added to the part and displayed in the assembly. The origin of the temporary body is relative to the part instead of the assembly. As a work-around, insert an interim part in the assembly with the origin aligned with the assembly and add the temporary body to the interim part.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

Check Method (IBody2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : Check Method (IBody2)

Obsolete. Superseded by [IBody2::Check3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Check() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Integer  
  
value = instance.Check()
```

C#

```
System.int Check()
```

C++/CLI

```
System.int Check();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Body2::Check](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Check2 Method (IBody2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : Check2 Method (IBody2)

Obsolete. Superseded by [IBody2::Check3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Check2() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Integer  
  
value = instance.Check2()
```

C#

```
System.int Check2()
```

C++/CLI

```
System.int Check2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Body2::Check2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Check3 Property (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : Check3 Property (IBody2)

Gets whether the body is a valid and returns an [IFaultEntity](#) object if any faults exist.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Check3 As FaultEntity
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As FaultEntity

value = instance.Check3
```

C#

```
FaultEntity Check3 {get;}
```

C++/CLI

```
property FaultEntity^ Check3 {
    FaultEntity^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pointer to [IFaultEntity](#) object

Visual Basic for Applications (VBA) Syntax

See [Body2::Check3](#).

Example

[Check Bodies for Faults \(VBA\)](#)
[Get and Fill Gaps in Body \(C#\)](#)
[Get and Fill Gaps in Body \(VB.NET\)](#)
[Get and Fill Gaps in Body \(VBA\)](#)

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IEdge::Check Property](#)
[IFace2::Check Property](#)

Availability

ConvertToMeshBody Method (IBody2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : ConvertToMeshBody Method (IBody2)

Converts a solid body to a mesh BREP (boundary representation) body.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function ConvertToMeshBody( _
    ByVal GroupFacetsIntoFaces As System.Boolean, _
    ByVal KeepOriginalBody As System.Boolean, _
    ByVal HideOriginalBody As System.Boolean, _
    ByVal MeshRefinement As System.Double, _
    ByVal AdvancedMeshRefinement As System.Boolean, _
    ByVal MaximumDistance As System.Double, _
    ByVal MaximumAngle As System.Double, _
    ByVal DefineMaxElementSize As System.Boolean, _
    ByVal ElementSize As System.Double, _
    ByRef ErrorCode As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim GroupFacetsIntoFaces As System.Boolean
Dim KeepOriginalBody As System.Boolean
Dim HideOriginalBody As System.Boolean
Dim MeshRefinement As System.Double
Dim AdvancedMeshRefinement As System.Boolean
Dim MaximumDistance As System.Double
Dim MaximumAngle As System.Double
Dim DefineMaxElementSize As System.Boolean
Dim ElementSize As System.Double
Dim ErrorCode As System.Integer
Dim value As System.Object
```

```
value = instance.ConvertToMeshBody(GroupFacetsIntoFaces, KeepOriginalBody, HideOriginalBody,
MeshRefinement, AdvancedMeshRefinement, MaximumDistance, MaximumAngle, DefineMaxElementSize,
ElementSize, ErrorCode)
```

C#

```
System.object ConvertToMeshBody(
    System.bool GroupFacetsIntoFaces,
    System.bool KeepOriginalBody,
    System.bool HideOriginalBody,
    System.double MeshRefinement,
    System.bool AdvancedMeshRefinement,
    System.double MaximumDistance,
    System.double MaximumAngle,
    System.bool DefineMaxElementSize,
    System.double ElementSize,
    out System.int ErrorCode
)
```

C++/CLI

```
System.Object^ ConvertToMeshBody(
    & System.bool GroupFacetsIntoFaces,
    & System.bool KeepOriginalBody,
    & System.bool HideOriginalBody,
    & System.double MeshRefinement,
    & System.bool AdvancedMeshRefinement,
    & System.double MaximumDistance,
```

```

& System.double MaximumAngle,
& System.bool DefineMaxElementSize,
& System.double ElementSize,
& [Out] System.int ErrorCode
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GroupFacetsIntoFaces

True to group facets into multiple faces, false to convert the mesh into a single face; valid only for graphics bodies

KeepOriginalBody

True to maintain a reference copy of the original solid or surface body; false to not

HideOriginalBody

True to hide the original body, false to not; valid only if KeepOriginalBody is true

MeshRefinement

0.0 (larger mesh facets) <= Percent refinement <= 100.0 (smaller mesh facets); valid only if AdvancedMeshRefinement is false

AdvancedMeshRefinement

True to specify maximum distance and angle, false to not

MaximumDistance

Maximum distance deviation in meters

MaximumAngle

Maximum angle deviation in radians

DefineMaxElementSize

True to specify the maximum length of the fins, false to not

ElementSize

Length of fins in meters; valid only if DefineMaxElementSize is true

ErrorCode

0 if successful, -1 if errors

Return Value

Array of:

One [body](#) (if this is a solid body)

- or -

Multiple surface bodies (if this is a graphics body)

Visual Basic for Applications (VBA) Syntax

See [Body2::ConvertToMeshBody](#).

Example

'VBA

'=====

'Preconditions:

- '1. Open a part that contains a solid body, a graphics body, or a mesh body.
- '2. If you open a solid body, this macro converts it to a mesh body.
- '3. If you open a graphics body that is meshed or you open a mesh body, multi-select the mesh facets that
' can be used to create a plane, cylinder, cone, or sphere.
- '4. Open an Immediate window.

'Postconditions:

- '1. Inspect the Immediate window.
- '2. If a solid body was opened, **Body-Convert to Mesh BodyX** is created in the FeatureManager design tree.
- '3. If a graphics body or a mesh body was opened and you selected facets,
' this macro creates a **Surface-From-MeshX** feature from the selected mesh facets
' using the specified facet tolerance and extend surface size.
' If ErrorCodeSurfaceFromMesh is -1, then select more mesh facets and/or
' change the IGraphicsBody/IMeshBody::SurfaceFromMesh parameters (SurfaceType/FacetTolerance/ExtendSurfaceSize).

'=====

Option Explicit

Sub main()

```

Dim swApp As SldWorks.SldWorks
Dim swModelDoc As ModelDoc2
Dim swFeatMgr As FeatureManager
Dim swSelMgr As SelectionMgr
Dim swSelData As SelectData

```

```

Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long

```

```

Set swApp = Application.SldWorks
Set swModelDoc = swApp.ActiveDoc
Set swSelMgr = swModelDoc.SelectionManager
Set swSelData = swSelMgr.CreateSelectData
swSelData.Mark = 1

```

```

Dim swPartDoc As PartDoc
Set swPartDoc = swModelDoc

```

```

Dim vAllBodies As Variant
vAllBodies = swPartDoc.GetBodies2(swAllBodies, False)
Dim swGenericBody As Body2
Dim IBodyType As Long

Dim swBody As Body2
Dim swGraphicsBody As GraphicsBody
Dim swMeshBody As MeshBody
Dim swBodyFeat As Feature

'Use first body that is Solid, Mesh or Graphics Body
Set swBody = Nothing
Set swMeshBody = Nothing
Set swGraphicsBody = Nothing
Dim i As Integer
For i = 0 To UBound(vAllBodies)
    Set swGenericBody = vAllBodies(i)
    IBodyType = swGenericBody.GetType
    If IBodyType = swBodyType_e.swSolidBody Then
        Debug.Print "Body type is swSolidBody"
        Set swBody = swGenericBody
        Exit For
    ElseIf IBodyType = swBodyType_e.swMeshBody Then
        Debug.Print "Body type is swMeshBody"
        Set swBody = swGenericBody
        Set swMeshBody = swBody.GetMeshBody
        Exit For
    ElseIf IBodyType = swBodyType_e.swGraphicsBody Then
        Debug.Print "Body type is swGraphicsBody"
        Set swBody = swGenericBody
        Set swGraphicsBody = swBody.GetGraphicsBody
        Exit For
    End If
Next i

Dim ISelObjType As Long
Dim ISelObjCount As Long
ISelObjType = swSelMgr.GetSelectedObjectType3(1, -1)

If ISelObjType = swSelNOTHING Then
    Debug.Print "Nothing is selected"
ElseIf ISelObjType = swSelSOLIDBODIES Then
    Debug.Print "swSolidBody is selected"
ElseIf ISelObjType = swSelGRAPHICSBODY Then
    Debug.Print "swGraphicsBody is selected"
ElseIf ISelObjType = swSelMESHESOLIDBODIES Then
    Debug.Print "swMeshBody is selected"
ElseIf ISelObjType = swSelFACETS Then
    Debug.Print "swSelFACETS is selected"
Else
    Debug.Print "Nothing interesting is selected"
End If

Dim swSurfaceBody As Body2
Dim SurfaceType As Long
Dim FacetTolerance As Double
Dim ExtendSurfaceSize As Double
Dim ErrorCodeSurfaceFromMesh As Long
Set swSurfaceBody = Nothing
SurfaceType = swSurfaceTypes_e.PLANE_TYPE
FacetTolerance = 0
ExtendSurfaceSize = 0.023
ErrorCodeSurfaceFromMesh = 0

If Not swGraphicsBody Is Nothing Then 'And ISelObjType = swSelFACETS
    Debug.Print "Testing GraphicsBody.SurfaceFromMesh"
    Set swSurfaceBody = swGraphicsBody.SurfaceFromMesh(SurfaceType, FacetTolerance, ExtendSurfaceSize,
    ErrorCodeSurfaceFromMesh)
    Debug.Print "ErrorCodeSurfaceFromMesh = " & ErrorCodeSurfaceFromMesh
ElseIf Not swMeshBody Is Nothing Then 'And ISelObjType = swSelFACETS
    Debug.Print "Testing MeshBody.SurfaceFromMesh"
    Set swSurfaceBody = swMeshBody.SurfaceFromMesh(SurfaceType, FacetTolerance, ExtendSurfaceSize, ErrorCodeSurfaceFromMesh)
    Debug.Print "ErrorCodeSurfaceFromMesh = " & ErrorCodeSurfaceFromMesh
ElseIf Not swBody Is Nothing Then
    Debug.Print "Testing ConvertToMeshBody"
    swBody.Select2 False, swSelData
    Dim outBodiesArray As Variant
    Dim GroupFacetsIntoFaces As Boolean
    Dim KeepOriginalBody As Boolean
    Dim HideOriginalBody As Boolean
    Dim MeshRefinement As Double
    Dim AdvancedMeshRefinement As Boolean

```

```

Dim MaximumDistance As Double
Dim MaximumAngle As Double
Dim DefineMaxElementSize As Boolean
Dim ElementSize As Double
Dim ErrorCode As Long
GroupFacetsIntoFaces = False
KeepOriginalBody = False
HideOriginalBody = False
MeshRefinement = 0.005
AdvancedMeshRefinement = False
MaximumDistance = 0.01
MaximumAngle = 0.05
DefineMaxElementSize = False
ElementSize = 0.01
ErrorCode = -1
outBodiesArray = swBody.ConvertToMeshBody(GroupFacetsIntoFaces, KeepOriginalBody, HideOriginalBody, MeshRefinement,
AdvancedMeshRefinement, MaximumDistance, MaximumAngle, DefineMaxElementSize, ElementSize, ErrorCode)
If IsEmpty(outBodiesArray) Then
    Debug.Print "outBodiesArray is Empty"
Else
    Dim ICount As Long
    ICount = UBound(outBodiesArray) - LBound(outBodiesArray) + 1
    If ICount > 0 Then
        Debug.Print "outBodiesArray has " & ICount & " elements."
        Dim swNewBody As Body2
        Set swNewBody = outBodiesArray(0)
        If Not swNewBody Is Nothing Then
            Debug.Print "Returned a Body"
        Else
            Debug.Print "No Body Returned"
        End If
    Else
        Debug.Print "outBodiesArray has no elements."
    End If
End If
Debug.Print "ErrorCodeConvertToMesh = " & ErrorCode
End If

End Sub

```

Remarks

Call this method for each body you want to convert.

This method creates **Body-Convert to Mesh Body** in the FeatureManager design tree.

This method is analogous to the **SOLIDWORKS** menu **Insert > Mesh > Convert To Mesh Body**. For more information, see the **SOLIDWORKS Design user-interface help > Parts and Features > Graphics Mesh and Mesh BREP Bodies > Converting Solid, Surface or Graphics Bodies to Mesh** topic.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

[SOLIDWORKS API Help](#)

[Send comments on this topic.](#)

Copy Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : Copy Method (IBody2)

Obsolete. Superseded by [IBody2::Copy2](#)

.NET Syntax

Visual Basic (Declaration)

Function Copy() As System.Object

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Object  
  
value = instance.Copy()
```

C#

```
System.Object Copy()
```

C++/CLI

```
System.Object^ Copy();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Object for the copied body

Visual Basic for Applications (VBA) Syntax

See [Body2::Copy](#).

Example

[Change Imported File Associated with an Imported Feature \(VBA\)](#)
[Combine Assembly Components into Part \(VBA\)](#)
[Get Differences Between Parts \(VBA\)](#)
[Delete Faces \(VBA\)](#)
[Save Rollbacked Part as a Parasolid File \(VBA\)](#)
[Create and Convert Non-manifold Bodies \(C#\)](#)
[Create and Convert Non-manifold Bodies \(VB.NET\)](#)
[Create and Convert Non-manifold Bodies \(VBA\)](#)
[Delete Blended Faces \(C#\)](#)
[Delete Blended Faces \(VB.NET\)](#)
[Delete Blended Faces \(VBA\)](#)

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICopy Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

Copy2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : Copy2 Method (IBody2)

Gets a copy of this body.

.NET Syntax

Visual Basic (Declaration)

```
Function Copy2( _  
    ByVal PreserveFaceIDs As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim PreserveFaceIDs As System.Boolean  
Dim value As System.Object  
  
value = instance.Copy2(PreserveFaceIDs)
```

C#

```
System.Object Copy2(  
    System.Boolean PreserveFaceIDs  
)
```

C++/CLI

```
System.Object^ Copy2(  
& System.Boolean PreserveFaceIDs  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PreserveFaceIDs
True to copy face IDs, false to create new IDs

Return Value

Copied IBody2

Visual Basic for Applications (VBA) Syntax

See [Body2::Copy2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateBaseFeature Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateBaseFeature Method (IBody2)

Creates a base feature for the imported body.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBaseFeature( _  
    ByVal BodyIn As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim BodyIn As System.Object  
Dim value As System.Boolean  
  
value = instance.CreateBaseFeature(BodyIn)
```

C#

```
System.Boolean CreateBaseFeature(  
    System.Object BodyIn  
)
```

C++/CLI

```
System.Boolean CreateBaseFeature(  
& System.Object^ BodyIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyIn

[Body](#) on which to create base feature

Return Value

True if the base feature was created, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateBaseFeature](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICreateBaseFeature Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateBlendSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateBlendSurface Method (IBody2)

Creates a constant radius rolling-ball blend surface (also known as a pipe surface) between two side surfaces.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBlendSurface( _  
    ByVal Surface1 As System.Object, _  
    ByVal Range1 As System.Double, _  
    ByVal Surface2 As System.Object, _  
    ByVal Range2 As System.Double, _  
    ByVal StartVec As System.Object, _  
    ByVal EndVec As System.Object, _  
    ByVal HaveHelpVec As System.Integer, _  
    ByVal HelpVec As System.Object, _  
    ByVal HaveHelpBox As System.Integer, _  
    ByVal HelpBox As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Surface1 As System.Object  
Dim Range1 As System.Double  
Dim Surface2 As System.Object  
Dim Range2 As System.Double  
Dim StartVec As System.Object  
Dim EndVec As System.Object  
Dim HaveHelpVec As System.Integer  
Dim HelpVec As System.Object  
Dim HaveHelpBox As System.Integer  
Dim HelpBox As System.Object  
Dim value As System.Object
```

```
value = instance.CreateBlendSurface(Surface1, Range1, Surface2, Range2, StartVec, EndVec,  
HaveHelpVec, HelpVec, HaveHelpBox, HelpBox)
```

C#

```
System.Object CreateBlendSurface(
    System.Object Surface1,
    System.Double Range1,
    System.Object Surface2,
    System.Double Range2,
    System.Object StartVec,
    System.Object EndVec,
    System.Int HaveHelpVec,
    System.Object HelpVec,
    System.Int HaveHelpBox,
    System.Object HelpBox
)
```

C++/CLI

```
System.Object^ CreateBlendSurface(
    & System.Object^ Surface1,
    & System.Double Range1,
    & System.Object^ Surface2,
    & System.Double Range2,
    & System.Object^ StartVec,
    & System.Object^ EndVec,
    & System.Int HaveHelpVec,
    & System.Object^ HelpVec,
    & System.Int HaveHelpBox,
    & System.Object^ HelpBox
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Surface1

First side surface

Range1

Signed offset of Surface1

Surface2

Second side surface

Range2

Signed offset of Surface2

StartVec

Array of 3 doubles representing a point close to the start of the blend spine

EndVec

Array of 3 doubles representing a point close to the end of the blend spine

HaveHelpVec

Optional. True if help vector is provided

HelpVec

Array of 3 doubles representing the direction of the Help vector

HaveHelpBox

Optional. True if box is provided

HelpBox

Array of 6 doubles bounding the area of interest

Return Value

Newly created surface

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateBlendSurface](#).

Remarks

The surface generated by this method is the path traced out when a sphere of the blend radius is rolled in contact with the two side surfaces. The center of the sphere rolls along the spine (the intersection of these two surfaces) when offset by the blend radius.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICreateBlendSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateBodyFromFaces Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateBodyFromFaces Method (IBody2)

Creates a temporary body from the faces.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBodyFromFaces( _  
    ByVal NumOfFaces As System.Integer, _  
    ByVal FaceList As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim NumOfFaces As System.Integer  
Dim FaceList As System.Object  
Dim value As System.Object  
  
value = instance.CreateBodyFromFaces(NumOfFaces, FaceList)
```

C#

```
System.object CreateBodyFromFaces(  
    System.int NumOfFaces,  
    System.object FaceList  
)
```

C++/CLI

```
System.Object^ CreateBodyFromFaces(  
& System.int NumOfFaces,  
& System.Object^ FaceList  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces

Number of faces to use to create the body

FaceList

Array containing the faces to use to create the body

Return Value

Object for the body

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateBodyFromFaces](#).

Example

[Create Body from Selected Faces \(C#\)](#)
[Create Body from Selected Faces \(VB.NET\)](#)
[Create Body from Selected Faces \(VBA\)](#)

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICreateBodyFromFaces Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateBodyFromSurfaces Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateBodyFromSurfaces Method (IBody2)

Creates a body from a list of trimmed surfaces.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBodyFromSurfaces() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Boolean
```

```
value = instance.CreateBodyFromSurfaces()
```

C#

```
System.bool CreateBodyFromSurfaces()
```

C++/CLI

```
System.bool CreateBodyFromSurfaces();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if solid body is created

False if:

- solid body is not created
- or -
- surface body is created

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateBodyFromSurfaces](#).

Example

[Create Body Using Trimmed Surfaces \(VBA\)](#)
[Create Imported Surface Body from Sketch \(C#\)](#)
[Create Imported Solid Body \(C#\)](#)
[Create Imported Solid Body \(VB.NET\)](#)
[Create Imported Solid Body \(VBA\)](#)

Remarks

To create a body using trimmed surfaces:

1. Create a new temporary body in a part using [IPartDoc::CreateNewBody](#).
2. Create the trimmed surfaces in the shape of the new body (for example, six square surfaces that intersect at the edges to form a cube).
 - Create a planar surface based on a root point and normal (two, three cell VARIANT arrays) using IBody2::CreatePlanarSurface(RootPoint, Normal).
 - Add a trimming loop to the planar surface using ISurface::AddTrimmingLoop2(Numcurves, _

Order, _
Dimen, _
Periodic, _
NumKnots, _
NumCtrlPoints, _
Knots, _
CtrlPointDbls, _
UVRange)

- Create a trimmed surface on the body based on the trimming loop that was just created. The arguments for Surface::AddTrimmingLoop2 and their values for a square are:

Argument	Description
NumCurves	Number of curves that make up the loop (4 Long)
Order	Orders for the spline curves ({2, 2, 2, 2} Array of Longs)
Dimen	Dimension of the control points for the spline curves ({2, 2, 2, 2} Array of Longs)
Periodic	Periodicity of the spline curves ({0, 0, 0, 0} Array of Longs)
NumKnots	Number of Knots of the spline curves ({4, 4, 4, 4} Array of Longs)
NumCtrlPoints	Number of Control points for the spline curves ({2, 2, 2, 2} Array of Longs)
Knots	Describes the locations of the knots ({0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1} Array of Doubles. Each knot represented by four numbers 0, 0, 1, 1)
CtrlPointDbls	Control points for the TrimmingLoop ({0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0} Array of Doubles. Describes the corners of the square)
UVRange	Min and max for the U and V values ({0, 1, 0, 1} Array of Doubles)

3. Sew the surfaces together into a new body using this method.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateBoundedSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateBoundedSurface Method (IBody2)

Creates a bounded surface from an independent base surface.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBoundedSurface( _  
    ByVal UOpt As System.Boolean, _  
    ByVal VOpt As System.Boolean, _  
    ByVal UvParams As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IBody2  
Dim UOpt As System.Boolean  
Dim VOpt As System.Boolean  
Dim UvParams As System.Object  
Dim value As System.Boolean
```



```
value = instance.CreateBoundedSurface(UOpt, VOpt, UvParams)
```

C#

```
System.bool CreateBoundedSurface(  
    System.bool UOpt,  
    System.bool VOpt,  
    System.object UvParams  
)
```

C++/CLI

```
System.bool CreateBoundedSurface(  
& System.bool UOpt,  
& System.bool VOpt,  
& System.Object^ UvParams  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UOpt

True if U parameter range is given in uvData, false for the entire U parameter range

VOpt

True if V parameter range is given in uvData, false for the entire V parameter range

UvParams

Array of 4 doubles (see **Remarks**)

Return Value

True if bounded surface creation was successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateBoundedSurface](#).

Remarks

Before you use this method, you must call one of the base surface creation methods, such as [IBody2::CreateBsplineSurface](#).

The UOpt and VOpt arguments allow the application to supply parameter range information so that a bounded surface can be made up from part of the base surface. If UOpt and VOpt are both set to false, then SOLIDWORKS uses the entire parameter ranges. This method fails for surface with infinite parameter ranges.

UvParams contains 4 doubles describing the UV parameter ranges.

U parameter range	<i>uvData[0]</i> and <i>uvData[1]</i> ; <i>UvData[0]</i> must be less than <i>uvData[1]</i>
V parameter range	<i>uvData[2]</i> and <i>uvData[3]</i> ; <i>UvData[2]</i> must be less than <i>uvData[3]</i>

To construct a solid body from bounded surfaces, call [IPartDoc::CreateNewBody](#) first, which arranges for a placeholder for this bounded surface

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICreateBoundedSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateBsplineSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateBsplineSurface Method (IBody2)

Creates a new B-spline surface.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function CreateBsplineSurface( _  
    ByVal Props As System.Object, _  
    ByVal UKnots As System.Object, _  
    ByVal VKnots As System.Object, _  
    ByVal CtrlPtCoords As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Props As System.Object  
Dim UKnots As System.Object  
Dim VKnots As System.Object  
Dim CtrlPtCoords As System.Object  
Dim value As System.Object  
  
value = instance.CreateBsplineSurface(Props, UKnots, VKnots, CtrlPtCoords)
```

C#

```
System.object CreateBsplineSurface(  
    System.object Props,  
    System.object UKnots,  
    System.object VKnots,  
    System.object CtrlPtCoords  
)
```

C++/CLI

```
System.Object^ CreateBsplineSurface(  
& System.Object^ Props,  
& System.Object^ UKnots,  
& System.Object^ VKnots,  
& System.Object^ CtrlPtCoords  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Props

Array containing 8 integers packed as 4 double elements (see **Remarks**)

UKnots

Array of numUKnots (see **Remarks**)

VKnots

Array of numVKnots (see **Remarks**)

CtrlPtCoords

Array of numCtrlPtCoord (see **Remarks**)

Return Value

A new B-spline surface

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateBsplineSurface](#).

Remarks

You can use this method with trimming curve creation routines (for example, [ISurface::AddTrimmingLoop2](#)) to construct a trimmed surface.

The Props array contains the following elements. These are integers packed into a double array in the Dispatch version.

- Uorder, Vorder
- numV_CtrlPoints, numU_CtrlPoints
- Uperiodicity, Vperiodicity

- DimensionControlPoints, UNUSED (set =0)

The number of elements in the UKnots array is given by:

$$\text{numUKnots} = \text{numU_CtrlPoints} + \text{Uorder}$$

The number of elements in the VKnots array is given by:

$$\text{numVKnots} = \text{numV_CtrlPoints} + \text{Vorder}$$

The number of elements in the ctrlPtCoords array is given by:

$$\text{numCtrlPtCoord} = (\text{NumV_CtrlPoints} * \text{NumU_CtrlPoints} * \text{DimensionControlPoints})$$

NOTE: The order of the UV control points are reversed if you used [ISurface::GetBSurfParams3](#) or [ISurface::IGetBSurfParams](#) to get the data. You do not need to reverse the UV control points; instead, you can switch the UV knots and related parameters.

For periodics, the first knot must have multiplicity = 1 and all excess multiplicity must be imposed on the last knot. Therefore, a valid periodic knot vector would be { 0, 1, 2, 3, 3, 3 } for a cubic curve. The control point on the seam of the closed curve cannot be replicated at both ends; it should occur only at the beginning.

To convert from a clamped periodic curve (numKnots = numCtrlPts + Order, ctrlPts replicated, knot multiplicity = order at each end) to a SOLIDWORKS periodic curve, remove all but one of the knots at the beginning of the vector and remove one at the end. Also remove the last control point to avoid the point replication.

Any existing object created by this method is destroyed if you call this method again.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICreateBsplineSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateExtrusionSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateExtrusionSurface Method (IBody2)

Creates a new surface of extrusion (infinitely long tabulated cylinder).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateExtrusionSurface( _
    ByVal ProfileCurve As System.Object, _
    ByVal AxisDirection As System.Object _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim ProfileCurve As System.Object
Dim AxisDirection As System.Object
Dim value As System.Object
```

```
value = instance.CreateExtrusionSurface(ProfileCurve, AxisDirection)
```

C#

```
System.object CreateExtrusionSurface(
    System.object ProfileCurve,
    System.object AxisDirection
)
```

C++/CLI

```
System.Object^ CreateExtrusionSurface(  
& System.Object^ ProfileCurve,  
& System.Object^ AxisDirection  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ProfileCurve

Profile curve

AxisDirection

Array of 3 doubles (x,y,z)

Return Value

New surface of extrusion (tabulated cylinder)

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateExtrusionSurface](#).

Remarks

You can use this method with:

- A set of related functions that construct a body from trimmed surfaces. The profile curve is extruded along the direction vector of axis direction, the new surface being the envelope of the curve. The profile curve must be of type line, circle, or B-spline curve.
- Trimming curve creation routines (for example [ISurface::AddTrimmingLoop2](#)) to construct a trimmed tabulated cylinder.

Any existing object created by this method is destroyed if you call this method again.

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

[IBody2::ICreateExtrusionSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateNewSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateNewSurface Method (IBody2)

Creates a handle for a new surface to serve as geometry for a face to be added to the body.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateNewSurface() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
```

```
Dim value As System.Object
```

```
value = instance.CreateNewSurface()
```

C#

```
System.object CreateNewSurface()
```

C++/CLI

```
System.Object^ CreateNewSurface();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

New surface

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateNewSurface](#).

Remarks

This method is the first in a set of related functions that construct a body from trimmed surfaces. Internally, this function also creates a list that SOLIDWORKS uses as a place-holder for trimming curves used to trim the surface to the desired shape.

Any existing object created by this method is destroyed if you call this method again.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICreateNewSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateOffsetSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateOffsetSurface Method (IBody2)

Creates a new surface offset from an existing surface.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateOffsetSurface(  
    ByVal SurfaceIn As System.Object, _  
    ByVal Distance As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim SurfaceIn As System.Object  
Dim Distance As System.Double  
Dim value As System.Object
```

```
value = instance.CreateOffsetSurface(SurfaceIn, Distance)
```

C#

```
System.object CreateOffsetSurface(  
    System.object SurfaceIn,  
    System.double Distance  
)
```

C++/CLI

```
System.Object^ CreateOffsetSurface(  
& System.Object^ SurfaceIn,  
& System.double Distance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SurfaceIn

Surface from which to offset the new surface

Distance

Offset distance

Return Value

Newly created offset surface

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateOffsetSurface](#).

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

[IBody2::ICreateOffsetSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreatePlanarSurface Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreatePlanarSurface Method (IBody2)

Creates a new infinite planar surface.

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlanarSurface(  
    ByVal VRootPoint As System.Object, _  
    ByVal VNormal As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim VRootPoint As System.Object  
Dim VNormal As System.Object  
Dim value As System.Object
```

```
value = instance.CreatePlanarSurface(VRootPoint, VNormal)
```

C#

```
System.object CreatePlanarSurface(  
    System.object VRootPoint,  
    System.object VNormal  
)
```

C++/CLI

```
System.Object^ CreatePlanarSurface(  
& System.Object^ VRootPoint,  
& System.Object^ VNormal  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VRootPoint

Array of 3 doubles (x,y,z)

VNormal

Array of 3 doubles (x,y,z)

Return Value

New planar surface

Visual Basic for Applications (VBA) Syntax

See [Body2::CreatePlanarSurface](#).

Example

[Create Body Using Trimmed Surfaces \(VBA\)](#)

Remarks

You can use this method with:

- A set of related functions that construct a body from trimmed surfaces.
- Trimming curve creation routines (for example, [ISurface::AddTrimmingLoop2](#)) to construct a trimmed surface.

To create a planar trimmed surface directly from the vertex points, see [IBody2::CreatePlanarTrimSurfaceDLL](#).

Any existing object created by this method is destroyed if you call this method again.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::CreatePlanarSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreatePlanarTrimSurfaceDLL Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreatePlanarTrimSurfaceDLL Method (IBody2)

Creates a planar trim surface for this body.

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePlanarTrimSurfaceDLL( _  
    ByVal Points As System.Object, _  
    ByVal Normal As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Points As System.Object  
Dim Normal As System.Object  
Dim value As System.Boolean
```

```
value = instance.CreatePlanarTrimSurfaceDLL(Points, Normal)
```

C#

```
System.bool CreatePlanarTrimSurfaceDLL(  
    System.object Points,  
    System.object Normal  
)
```

C++/CLI

```
System.bool CreatePlanarTrimSurfaceDLL(  
& System.Object^ Points,  
& System.Object^ Normal  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Points

Array of the points for the surface; trim curves are automatically created between each sequential vertex

Normal

Array of normal vector for the surface

Return Value

True if planar trim surface is created, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::CreatePlanarTrimSurfaceDLL](#).

Example

[Create Imported Solid Body \(C#\)](#)
[Create Imported Solid Body \(VB.NET\)](#)
[Create Imported Solid Body \(VBA\)](#)

Remarks

You can use this method instead of [IBody2::CreateTrimmedSurface](#), [IBody2::ICreatePlanarSurface](#), and [ISurface::AddTrimmingLoop2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICreatePlanarTrimSurfaceDLL Method](#)

Availability

SOLIDWORKS 2004 SP1, Revision Number 12

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateRevolutionSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateRevolutionSurface Method (IBody2)

Creates a new surface of revolution.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateRevolutionSurface(  
    ByVal ProfileCurve As System.Object, _  
    ByVal AxisPoint As System.Object, _  
    ByVal AxisDirection As System.Object, _  
    ByVal ProfileEndPtParams As System.Object _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IBody2
Dim ProfileCurve As System.Object
Dim AxisPoint As System.Object
Dim AxisDirection As System.Object
Dim ProfileEndPtParams As System.Object
Dim value As System.Object

value = instance.CreateRevolutionSurface(ProfileCurve, AxisPoint, AxisDirection,
ProfileEndPtParams)
```

C#

```
System.object CreateRevolutionSurface(
    System.object ProfileCurve,
    System.object AxisPoint,
    System.object AxisDirection,
    System.object ProfileEndPtParams
)
```

C++/CLI

```
System.Object^ CreateRevolutionSurface(
    & System.Object^ ProfileCurve,
    & System.Object^ AxisPoint,
    & System.Object^ AxisDirection,
    & System.Object^ ProfileEndPtParams
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ProfileCurve
Profile curve (generatrix)
AxisPoint
Array of 3 doubles (x,y,z)
AxisDirection
Array of 3 doubles (x,y,z)
ProfileEndPtParams
Array of 2 doubles (uStart, uEnd) (see **Remarks**)

Return Value

New surface of revolution

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateRevolutionSurface](#).

Remarks

You can use this method with:

- A set of related functions that construct a body from trimmed surfaces.
- Trimming curve creation routines (for example, [ISurface::AddTrimmingLoop2](#)) to construct a trimmed surface of revolution.

If you pass ProfileEndPtParams to this method, the surface is trimmed in the axial direction; otherwise, it is infinite. SOLIDWORKS closes the surface periodic (period [0,2PI]) in the direction of revolution. The ProfileEndPtParams parameters indicate to SOLIDWORKS which part of the curve to spin. These parameters are used only when the profile curve intersects the revolve axis. You must pass the parameters in ascending order. SOLIDWORKS extends the curve from the given parameter range to meet the revolve axis and spins this portion of curve.

You can also pass an empty VARIANT object to ProfileEndPtParams; it cannot be NULL. You must define a variable of type VARIANT.

Any existing object created by this method is destroyed if you call this method again.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICreateRevolutionSurface Method](#)

Availability

CreateRuledSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateRuledSurface Method (IBody2)

Creates a ruled surface from the specified curves and apex point.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateRuledSurface( _
    ByVal Curve1 As System.Object, _
    ByVal Curve2 As System.Object, _
    ByVal ApexPoint As System.Object _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim Curve1 As System.Object
Dim Curve2 As System.Object
Dim ApexPoint As System.Object
Dim value As System.Object

value = instance.CreateRuledSurface(Curve1, Curve2, ApexPoint)
```

C#

```
System.object CreateRuledSurface(
    System.object Curve1,
    System.object Curve2,
    System.object ApexPoint
)
```

C++/CLI

```
System.Object^ CreateRuledSurface(
    & System.Object^ Curve1,
    & System.Object^ Curve2,
    & System.Object^ ApexPoint
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Curve1
First curve

Curve2
Second curve

ApexPoint
Array of 3 doubles (x, y, z), the apex point

Return Value

Pointer to ruled [surface](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateRuledSurface](#).

Remarks

Any existing object created by this method is destroyed if you call this method again.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICreateRuledSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateTempBodyFromSurfaces Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateTempBodyFromSurfaces Method (IBody2)

Creates a temporary body from a list of existing trimmed surfaces.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateTempBodyFromSurfaces() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Object

value = instance.CreateTempBodyFromSurfaces()
```

C#

```
System.Object CreateTempBodyFromSurfaces()
```

C++/CLI

```
System.Object^ CreateTempBodyFromSurfaces();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Body

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateTempBodyFromSurfaces](#).

Remarks

This method is the final call to a set of related functions (same as [IBody2::CreateBodyFromSurfaces](#)) that are designed to construct a temporary body from trimmed surfaces.

The first call in this process should be to [IPartDoc::CreateNewBody](#), so as to arrange for a placeholder for all the trimmed surfaces.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ICreateTempBodyFromSurfaces Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateTrimmedSurface Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : CreateTrimmedSurface Method (IBody2)

Creates a trimmed surface from a base surface and a list of existing trimming curves.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateTrimmedSurface() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Boolean

value = instance.CreateTrimmedSurface()
```

C#

```
System.bool CreateTrimmedSurface()
```

C++/CLI

```
System.bool CreateTrimmedSurface();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if trimmed surface creation is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::CreateTrimmedSurface](#).

Example

[Create Body Using Trimmed Surfaces \(VBA\)](#)
[Create Imported Surface Body from Sketch \(C#\)](#)

Remarks

Before you call this method, you must call one of the base-surface creation methods (such as [IBody2::CreatePlanarSurface](#) or [IBody2::!CreatePlanarSurface](#)) and the trimming-curve creation method [ISurface::AddTrimmingLoop2](#) or [ISurface::!AddTrimmingLoops](#).

This method serves as the final call in a set of related functions that are designed to construct a trimmed surface from a base surface (possibly infinite) and a set of trimming curves.

If you want to construct a solid body from trimmed surfaces, call [IPartDoc::CreateNewBody](#) or [IPartDoc::!CreateNewBody2](#) first, which arranges for a place-holder for this trimmed surface.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

DeleteBlends Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DeleteBlends Method (IBody2)

Obsolete. Superseded by [IBody2::DeleteBlends2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteBlends( _  
    ByVal NumOfFaces As System.Integer, _  
    ByVal FaceList As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim NumOfFaces As System.Integer  
Dim FaceList As System.Object  
Dim value As System.Boolean  
  
value = instance.DeleteBlends(NumOfFaces, FaceList)
```

C#

```
System.bool DeleteBlends(  
    System.int NumOfFaces,  
    System.object FaceList  
)
```

C++/CLI

```
System.bool DeleteBlends(  
& System.int NumOfFaces,  
& System.Object^ FaceList  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces

FaceList

Visual Basic for Applications (VBA) Syntax

See [Body2::DeleteBlends](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

DeleteBlends2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DeleteBlends2 Method (IBody2)

Obsolete. Superseded by [IBody2::DeleteBlends3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteBlends2( _  
    ByVal NumOfFaces As System.Integer, _  
    ByVal FaceList As System.Object, _  
    ByVal DoLocalCheck As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim NumOfFaces As System.Integer  
Dim FaceList As System.Object
```

```
Dim DoLocalCheck As System.Boolean
Dim value As System.Boolean

value = instance.DeleteBlends2(NumOfFaces, FaceList, DoLocalCheck)
```

C#

```
System.bool DeleteBlends2(
    System.int NumOfFaces,
    System.object FaceList,
    System.bool DoLocalCheck
)
```

C++/CLI

```
System.bool DeleteBlends2(
    & System.int NumOfFaces,
    & System.Object^ FaceList,
    & System.bool DoLocalCheck
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces
Number of faces to delete
FaceList
List of [faces](#) to delete
DoLocalCheck
True to perform a local check, false to not

Return Value

True if the set of fillet faces are removed, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2.DeleteBlends2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::DeleteBlends2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

DeleteBlends3 Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DeleteBlends3 Method (IBody2)

Removes a set of fillet faces from a temporary body and heals the body.

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteBlends3( _
    ByVal FaceList As System.Object, _
    ByVal DoLocalCheck As System.Boolean, _
    ByVal UsePlanarCap As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IBody2
Dim FaceList As System.Object
```



```
Dim DoLocalCheck As System.Boolean
Dim UsePlanarCap As System.Boolean
Dim value As System.Boolean

value = instance.DeleteBlends3(FaceList, DoLocalCheck, UsePlanarCap)
```

C#

```
System.bool DeleteBlends3(
    System.object FaceList,
    System.bool DoLocalCheck,
    System.bool UsePlanarCap,
)
```

C++/CLI

```
System.bool DeleteBlends3(
    & System.Object^ FaceList,
    & System.bool DoLocalCheck,
    & System.bool UsePlanarCap,
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceList

List of [faces](#) to delete

DoLocalCheck

True to perform a local check, false to not

UsePlanarCap

True to use planar caps, false to not (see **Remarks**)

Return Value

True if the set of fillet faces are removed, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::DeleteBlends3](#).

Example

[Delete Blended Faces \(C#\)](#)

[Delete Blended Faces \(VB.NET\)](#)

[Delete Blended Faces \(VBA\)](#)

Remarks

Typically when deleting blends, an entire chain of blends is deleted. However, if only a few blends are deleted from a chain of blends and the UsePlanarCap parameter is not set to true, then the resultant body might be invalid.

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

[IBody2::DeleteBlends3 Method](#)

[IBody5::DeleteFaces5 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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DeleteFaces Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DeleteFaces Method (IBody2)

Obsolete. Superseded by [IBody2::DeleteFaces3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteFaces( _  
    ByVal NumOfFaces As System.Integer, _  
    ByVal FaceList As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim NumOfFaces As System.Integer  
Dim FaceList As System.Object  
Dim value As System.Boolean  
  
value = instance.DeleteFaces(NumOfFaces, FaceList)
```

C#

```
System.bool DeleteFaces(  
    System.int NumOfFaces,  
    System.object FaceList  
)
```

C++/CLI

```
System.bool DeleteFaces(  
& System.int NumOfFaces,  
& System.Object^ FaceList  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces

FaceList

Visual Basic for Applications (VBA) Syntax

See [Body2::DeleteFaces](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)